

Model Performances Evaluated for Infinite Dilution Activity Coefficients Prediction at 298.15 K

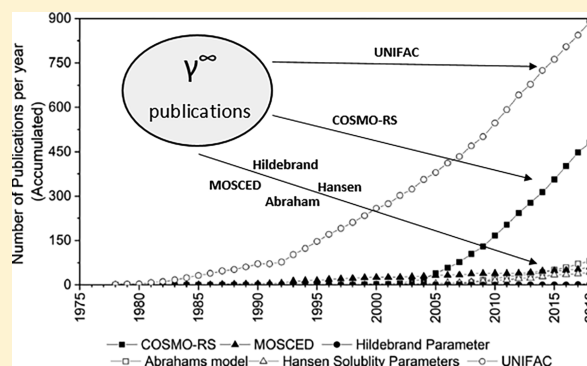
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Supporting Information

ABSTRACT: The infinite dilution activity coefficient (γ_i^∞) is often applied to characterize solvent–solute interactions, and, when accurately predicted, it can also serve as an early-stage solvent selection tool. Ample data are available on the use of a variety of models, which complicates decision making on which model to apply and when to apply it. A comparative study was performed for eight predictive models at 298.15 K, including the Hildebrand parameter and the Hansen solubility parameters. Also, three group contribution methods based on UNIFAC, COSMO-RS, the Abraham model, and the MOSCED model were evaluated. Overall, the MOSCED model and the Abraham model are most accurate for molecular solvents and ionic liquids, respectively, with average relative deviations of $16.2\% \pm 1.35\%$ and $65.1\% \pm 4.50\%$.

Therefore, cautious decision making based on predicted γ_i^∞ in ionic liquids should always be done, because of the expected significant deviations. A MOSCED model for ionic liquids could be a potential approach for higher accuracy.



1. INTRODUCTION

The global community relies on the chemical industry for the production of goods from complex raw materials, such as oil and biomass. The separation processes required in these production routes account for up to 50% of the total energy costs in refineries,¹ and improving the efficiency of separations can significantly reduce the environmental impact of the chemical industry.² This can only be achieved when the separation processes are understood on the molecular level, which includes a good description of thermodynamic equilibria. An accurate description of these equilibria is possible with models such as UNIQUAC and NRTL, but requires labor-intensive experimental data. For initial stages of process design, including solvent selection and/or design, less-labor-intensive approaches for understanding intermolecular interactions are desired. A range of predictive models is available to provide engineers with first estimates for (inter)molecular behavior.^{3–11}

An appropriate indicator of intermolecular interactions underlying thermodynamic equilibria is the infinite dilution activity coefficient.¹² Activity coefficients describe the thermodynamic nonideality between two substances due to intermolecular interactions. These intermolecular interactions are induced by van der Waals interactions^{13–15} and electrostatic interactions, such as intermolecular or intramolecular hydrogen bonding effects,¹⁶ consequently causing either positive or negative deviation from Raoult's law. Net attractive interactions result in an activity coefficient (γ) below unity, and net repulsive interactions result in a γ above unity. Activity

coefficients are composition-dependent and a limiting case is where a solute is infinitely diluted in a solvent. The nonideal behavior of the solute at infinite dilution is solely induced by solvent–solute interaction, i.e., the effect of the molecular properties of the solvent on the activity coefficient of the solute. The activity coefficient in this limiting case is called the infinite dilution activity coefficient (γ_i^∞).¹²

Various methods are in use to estimate activity coefficients. Eight models were evaluated, as these are most commonly used. Seven of them having many similarities and one has a completely separate theoretical framework. Three solvation models (SMs) are chosen, which are, in order of increasing complexity, the Hildebrand parameter, the Hansen solubility parameter (HSP), and the Modified Separation of Cohesive Energy Density (MOSCED) model. These models attempt to describe the intermolecular interaction strength by increasingly more molecular parameters. The group contribution methods (GCMs) are similar in form to SMs, but differ in origin. GCMs attempt to describe the intermolecular interactions by empirical fitting of binary interaction coefficients of segments of the interacting molecules. The three GCMs that are included in this work are the original Universal Quasichemical Functional-group Activity Coefficients (UNIFAC) method, and two modifications thereof (Lyngby and Dortmund).

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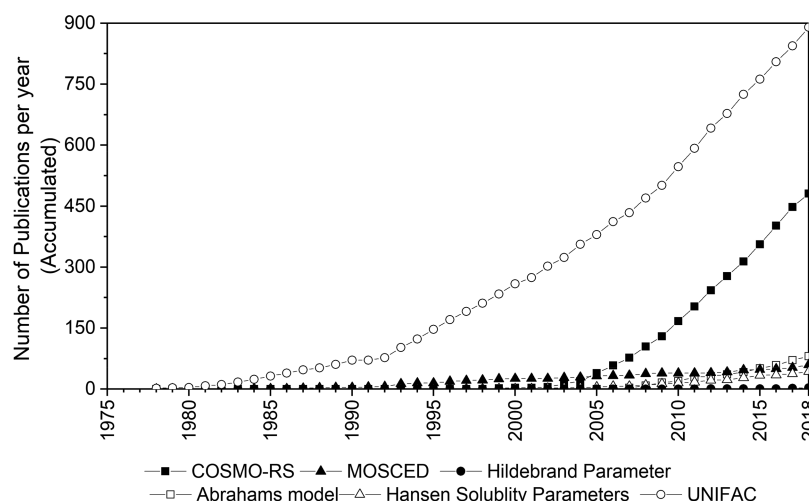


Figure 1. Number of publications where both the model and activity coefficient is mentioned extracted from Scopus. The search terms where “model” AND “infinite dilution activity coefficient” retrieved on December 3, 2018.

Despite these differences, all of these models still use the same entropic formulations. Next to these, a software package called the Conductor-like screening model for realistic solvents (COSMO-RS) is also evaluated. COSMO-RS has an entirely different theoretical framework and does not require any input parameters from the user besides the molecular structures. Similar to GCMs, COSMO-RS also divides molecules in segments. COSMO-RS divides the surface of molecules, as calculated by the quantum chemical density functional theory (DFT) approach in segments, and calculates the interaction energy for each segment. The molecular properties are then calculated by taking the integral over all segments.

GCMs such as variations on UNIFAC, and COSMO-RS are most commonly applied, as can be seen in Figure 1. However, this does not imply that these models are always most accurate in predicting the γ_i^∞ . Because the existing literature^{6,17–24} is inconclusive, the aim of this contribution is to extensively compare the performance of all these fundamentally different approaches for prediction of the γ_i^∞ of (a)polar solutes in (a)polar solvents. The performances of all evaluated models in predictions for a variety of chemical systems were evaluated and explained on the basis of their fundamental assumptions. Examples of such assumptions include that the entropy of the system does not differ from the ideal entropy, that the volume of the molecule does not change in a changing environment, or that there is no distinction between hydrogen-bond-accepting and hydrogen-bond-donating molecules.

The extensive evaluation of model performances yielded insight in the applicability of the models for systems with variations in intermolecular interactions and which models give the most accurate description of γ_i^∞ at 298.15 K. A heuristic approach for the model choice is given for all binary combinations of solute and solvent classes, e.g., apolar compounds, aromatic compounds, halogenated compounds, polar aprotic compounds, and polar protic compounds.

2. THEORETICAL FRAMEWORK

In this section, all of the models that were compared for prediction of γ_i^∞ are described. The models can be categorized into solvation models, GCMs, linear solvation energy relations, and COSMO-RS predictions that are of statistical thermodynamic nature. Because both the solvation models and the

GCMs make use of combinatorial and residual contributions that find their origin in the Flory–Huggins model; this model and the variations thereof are first discussed.

2.1. Flory–Huggins-Based Models. Nonideal behavior in mixtures can be induced by intermolecular interactions such as polarity, as well as by shape differences.¹⁶ The simplest cause of nonideal behavior is due to shape differences without polarity differences. This situation occurs in alkane mixtures for which activity coefficients can be described by the Flory–Huggins theory (eq 1), where the combinatorial contribution to the activity coefficient is formulated solely in terms of molecular volume differences.^{25,26}

$$\ln \gamma_i^c = \ln \left(\frac{\Phi_i}{x_i} \right) + 1 - \frac{\Phi_i}{x_i} \quad (1)$$

where Φ_i and x_i are the volume fraction and molar fraction of compound i , respectively.

The Flory–Huggins combinatorial approach assumes a very large number of nearest-neighbor sites, hence ignoring the fact that neighboring sites can be occupied by a segment of the same molecule. Consequently, the Flory–Huggins correlation overestimates the combinatorial contribution.²⁷ The Stavermann–Guggenheim modification attempts to correct this by incorporating the probability of vacant sites for polymer segments, although the combinatorial term is still overestimated, because the coordination number of all molecules is set.²⁸ The Kikic modifications attempted to correct the correlation by adding an exponent to the number of lattice sites,²⁹ but this resulted in an underprediction of the combinatorial term. Recently, Krooshof et al.²⁸ generalized the approach of Guggenheim and set loose the fixed coordination number. For this approach, the coordination number of all molecules in the system must be additionally specified.

When there is also a difference in the polarity of the molecules in the mixture, a residual correction can be added. This can be described by the Flory–Huggins free-energy parameter (χ_{12}), as equated in eq 2. The activity coefficient resulting from both combinatorial and residual terms is given in eq 3.

$$\ln \gamma_i^R = \chi_{12} \Phi_i^2 \quad (2)$$

$$\ln \gamma_i = \ln \gamma_i^c + \ln \gamma_i^R \quad (3)$$

Starting from eq 3, a wide range of models have been developed that vary in their formulation of γ_i^c and γ_i^R . In the next subsections, the most relevant solvation models and GCMs are described.

2.2. Solvation Models (SM). In an early attempt to describe and understand the strength of intermolecular interactions, Hildebrand⁷ defined the cohesive pressure or cohesive energy density (c) as the net result of the sum of all intermolecular interactions between the molecules. As a measurable quantity for the cohesive energy density in liquids below their boiling point, the molar vaporization energy (ΔU_{vap}) or enthalpy (ΔH_{vap}) is considered,³⁰ and correlated to the Hildebrand parameter (δ), according to eq 4.

$$\delta = \sqrt{c} = \sqrt{-\frac{\Delta U_{\text{vap}}}{V_m}} = \sqrt{\frac{\Delta H_{\text{vap}} - RT}{V_m}} \quad (4)$$

where V_m is the molar volume of the molecule i .

The cohesive pressure is the sum of all attractive interactions, which must be broken to vaporize the liquid, and large δ values are therefore obtained for highly polar substances and small δ values are obtained for weakly interacting substances (e.g., fluorocarbons).³⁰ Considering mixtures, the difference in δ of the constituents of that mixture can be interpreted as the difference in nature of these molecules and may be used as a measure for nonideality. In the Hildebrand–Scatchard equation (eq 5),⁷ the difference in δ is used in the expression for the Flory–Huggins free-energy parameter (χ_{12}):

$$\chi_{12} = \frac{V_m}{RT} (\delta - \delta')^2 \quad (5)$$

where δ and δ' are, respectively, the Hildebrand parameters of the solute and solvent.

Hansen⁸ proposed an extension of the Hildebrand parameter by separating the dispersion (δ_D), polar (δ_P), and hydrogen bonding (δ_{HB}) contribution. These three parameters are called the Hansen solubility parameters (HSP) and are linked to the Hildebrand parameter, as defined in eq 6.

$$\delta^2 = \delta_D^2 + \delta_P^2 + \delta_{HB}^2 \quad (6)$$

Often, the dispersive component was determined by homomorph methods, which estimates the dispersive parameter by evaluating an apolar molecule with almost the same size and shape of the polar compound.³⁰ The remainder can be subtracted from the Hildebrand parameter and split into the polar and hydrogen-bonding term. By optimizing the miscibility description an optimal split between both terms was chosen.³⁰

Common practice in using HSP is to plot the solute and solvents in a 3D Hansen space. The spatial distance between solute and solvent can be correlated to the solvation capability of the solvent, where shorter distances in the Hansen space allow better solubility. Hansen³¹ also suggested that the Flory–Huggins parameter can be determined using eq 7:

$$\chi_{12} = \alpha \frac{V_m}{RT} [(\delta_d - \delta'_d)^2 + 0.25(\delta_p - \delta'_p)^2 + 0.25(\delta_{HB} - \delta'_{HB})^2] \quad (7)$$

where, initially, α was taken to be 1, although Lindvig et al.⁹ showed that an α value of 0.6 increases the average accuracy of the model.

The Modified Separation of Cohesive Energy Density (MOSCED) model may be one of the most-extensive solvation models.²¹ In the MOSCED model, additional contributions to the Flory–Huggins parameter are considered that arise from significant variations in the cybotactic region due to the local organization, as a result of electrostatic interactions, such as hydrogen bonding. This local organization causes the geometric mean assumption not to be valid anymore for highly polar and associating compounds.²¹ To account for hydrogen bonding, the MOSCED model distinguishes acidic (α) and basic (β) contributions to hydrogen bonding. Similar to the HSP model, the summation of the terms results in the Hildebrand parameter, as can be seen in eq 8.³²

$$\delta^2 = \lambda^2 + \tau^2 + (\alpha\beta)^2 \quad (8)$$

where the dispersion constant λ and polarity constant τ are identical to the HSP parameters δ_d and δ_p , respectively. In addition, the interaction between induced dipoles is accounted for by the induction parameter, q . The model furthermore contains two empirical asymmetry factors: ψ and ξ . More information on the MOSCED parameters can be found in eqs S1–S8 in the Electronic Supporting Information (ESI).⁶ The resulting MOSCED-based equation for the Flory–Huggins parameter is given in eq 9:

$$\chi_{12} = \frac{V_m}{RT} \left[(\lambda - \lambda')^2 + \frac{q^2 q'^2 (\tau^T - \tau'^T)^2}{\psi} + \frac{(\alpha^T - \alpha'^T)(\beta^T - \beta'^T)}{\xi_1} \right] \quad (9)$$

2.3. Group Contribution Methods (GCMs). Also, the GCMs of UNIFAC and modifications thereof are the sum of a combinatorial part and a residual part. Each GCM model uses a different description for the combinatorial term. The UNIFAC and the modified UNIFAC(Do) models use the Guggenheim–Stavermann term³³ (eqs 10 and 11), while the modified UNIFAC(Ly) uses the Kikic modification,²⁹ as given in eq 12.

UNIFAC:

$$\ln \gamma_i^c = \ln(\Phi_i) + 1 - \Phi_i - S q_i \left[1 - \frac{\Phi_i}{\theta_i} + \ln \left(\frac{\Phi_i}{\theta_i} \right) \right] \quad (10)$$

modified UNIFAC(Do):

$$\ln \gamma_i^c = \ln(\Phi'_i) + 1 - \Phi'_i - S q_i \left[1 - \frac{\Phi_i}{\theta_i} + \ln \left(\frac{\Phi_i}{\theta_i} \right) \right] \quad (11)$$

modified UNIFAC(Ly):

$$\ln \gamma_i^c = \ln(\Phi''_i) + 1 - \Phi''_i \quad (12)$$

where Φ , θ , and q are, respectively, the volume fraction, surface fraction, and the coordination number, which is often set at 10. The modified UNIFAC(Do) model also has a modified volume fraction, on which more information is given in the ESI.

The residual contribution of all UNIFAC models is determined via eqs 13 and 14:

$$\ln \gamma_i^R = \sum_k \nu_k^{(i)} (\ln \Gamma_k - \ln \Gamma_k^{(i)}) \quad (13)$$

$$\ln \Gamma_k = Q_k \left(1 - \ln \left(\sum_m \theta_m \Psi_{mk} \right) - \sum_m \frac{\theta_m \Psi_{km}}{\sum_n \theta_n \Psi_{nm}} \right) \quad (14)$$

where Γ_k is the overall activity of moiety k , $\Gamma_k^{(i)}$ the activity of moiety k solely surrounded by moiety i , $\nu_k^{(i)}$ the occurrence of each moiety k in surrounded by moiety i , Q_k the van der Waals surface of group k , and Ψ the group binary interaction parameter.^{4,5,11}

2.4. Linear Solvation Energy Relationship (LSER).

Linearly combining solute and solvent descriptors was pioneered by Kamlet, Abboud, and Taft,^{34–38} and later Abraham³ introduced a now widely used LSER that consisted of five solute and five solvent descriptors (eq 15). An extension is made for ILs, where both the cation and anion have five unique solvent descriptors (see eq 16).³⁹ The Abraham model can be used to obtain either the gas-solvent partition coefficient (K_S) or the water-solvent partition coefficient (P_S).

$$\begin{aligned} \log(P_S) &= c + eE + sS + aA + bB + vV \\ \log(K_S) &= c + eE + sS + aA + bB + lL \end{aligned} \quad (15)$$

$$\begin{aligned} \log(P_S) &= c + (e_c + e_a)E + (s_c + s_a)S + (a_c + a_a)A \\ &\quad + (b_c + b_a)B + (v_c + v_a)V \\ \log(K_S) &= c + (e_c + e_a)E + (s_c + s_a)S + (a_c + a_a)A \\ &\quad + (b_c + b_a)B + (l_c + l_a)L \end{aligned} \quad (16)$$

The capital variables in eqs 15 and 16 are defined as the solute descriptors and the lowercase variables are the solvent descriptors. While the c -variable is a fitting constant, the latter terms are respectively the excess molar refraction [$\text{dm}^3 \text{mol}^{-1}/10$], the dipolarity/polarizability, hydrogen-bond acidity and basicity, the McGowan characteristic volume ($\text{dm}^3 \text{mol}^{-1}/100$), and the gas–hexadecane partition coefficient at 298.15 K. The descriptors describe the tendency to interact via σ - and π -electrons (e/E), the tendency to interact with (induced) multipole moments (s/S), the tendency to accept hydrogen bonds or donate electrons (A/b), the tendency to donate hydrogen bonds or accept electrons (a/B), and the tendency to either form (V/L) or the work required to form (v/l) cavities.

The determination of solute and solvent descriptors is done by multilinear regression of experimental γ_i^∞ data of either a set of solutes in a solvent or a solute in various solvents. The gas–solvent partition coefficient can consequently be linked to the γ_i^∞ parameter by eq 17:³⁹

$$\log(\gamma_i^\infty) = \log \left(\frac{RT}{P_j^\circ V_m} \right) - \log(K_S) \quad (17)$$

where P_j° and V_j are, respectively, the vapor pressure and molar volume of the pure solvent.

2.5. Conductor-like Screening Model for Real Solvents (COSMO-RS). The ab initio method developed by Klamt et al.,⁴⁰ called COSMO-RS, predicts chemical potentials, which can be used to calculate the value of γ_i^∞ . The first step is

always to perform quantum mechanical calculations to obtain the state of the geometrically optimized molecule. This step must be done only once, and the result can be stored in a database. The second step estimates the interaction energy of the optimized molecules with other molecules and can henceforth estimate molecular properties such as the γ_i^∞ parameter.⁴⁰

The energy of this state can be determined via dielectric continuum solvation methods. However, empirical parameters (e.g., atomic radii) are required to construct the molecular cavity within the conductor exterior. COSMO-RS implements element-specific radii, which are $\sim 17\%$ larger than van der Waals radii. The state of the molecule is consequently determined using any Self-Consistent Field (SCF) model, e.g., Hartree–Fock (HF) and density functional theory (DFT). Combining the COSMO approach with a SCF model results in not only the total energy of the molecule, but also the polarization charge density, or σ .⁴⁰

The σ -profile is “frozen” into place, while the conductor environment is “squeezed out”. A thin film of a conductor is left in place where different molecules will have an interface. The sum of the σ -value at the interface, ($\sigma + \sigma'$), is then the net polarization charge density. As the conductor is removed, the polarization charge density differences resemble intermolecular interactions with local contact energy, which may be described by

$$E_{\text{contact}}(\sigma, \sigma') = E_{\text{misfit}}(\sigma, \sigma') + E_{\text{hb}}(\sigma, \sigma') \quad (18)$$

An interaction distinction is made between the misfit energy (E_{misfit}) and the energy that is due to hydrogen bonding effects (E_{hb}). Both energy terms dissipate when the conjugant polarization charge densities are equal. If not, the misfit between both σ -values represents the electrostatic interaction energy between those segments. Additional interaction energy can be induced by two very polar surfaces with opposite signs via hydrogen bonding. The capability of COSMO-RS to predict a large number of chemical potentials of solutes in either a pure or mixed solvent enabled fast and versatile predictions regarding various equilibria and also γ_i^∞ .⁴⁰

3. METHODS

For all γ_i^∞ predictions, a systematic assessment was done at 298.15 K and all model specific parameters were imported from literature sources, as can be seen in the ESI. All simulations with COSMO-RS were performed with COSMOthermX C30_1705, in which a pure solvent phase was defined and the activity coefficient of the solute in that phase was estimated. Molecules that were not available in the databases were created with TurboMole TmoleX 3.4, using the TZVPD-Fine parametrization.

4. RESULTS

4.1. General Averaged Model Predictions. The accuracy of all models was determined by comparing the predictions with experimental values taken from the literature. An extensive overview of all experimental data is presented in the ESI. The accuracy of the models is evaluated by determining the average relative deviation (ARD) between the predicted and experimental γ_i^∞ values, as given in eq 19.

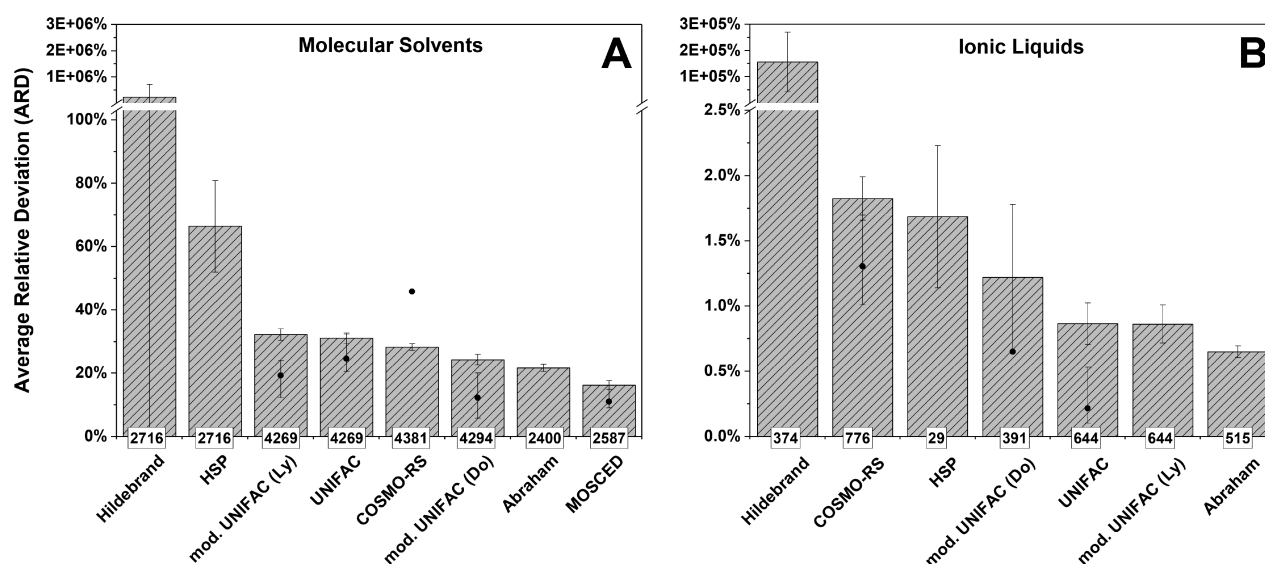


Figure 2. Evaluation of various predictive models for γ_i^∞ for (A) molecular solvents and (B) ionic liquids (ILs) at 298.15 K. On the y-axis, the ARD is presented within the boxes the total amount of comparisons made. The experimental γ_i^∞ is collected in the ESI. The integrated scatter plot depicts similar comparison made in the literature for various models, e.g., modified UNIFAC(Ly),^{17–19} UNIFAC,^{17–22,42,43} COSMO-RS,^{18,44} modified UNIFAC(Do),^{17,19,22,23,45–47} and MOSCED.^{6,20–22,24}

$$\text{ARD} = \frac{\sum_{i=1}^N \left| \frac{(\gamma_i^\infty)_{\text{model}} - (\gamma_i^\infty)_{\text{experimental}}}{(\gamma_i^\infty)_{\text{experimental}}} \right|}{N} \quad (19)$$

The ARD for prediction of γ_i^∞ for all solutes in molecular solvents and ILs by each model, and their 95% confidence intervals, are depicted in Figure 2. For molecular solvents, eight predictive models have been evaluated; for ILs, seven predictive models due to lack of MOSCED parameters for ILs have been evaluated. The ARD of the various models differs significantly, as can be seen in Figure 2. For both the molecular solvents and the ILs, the most inaccurate model is the Hildebrand model. Evidently, using only the evaporation enthalpy and the molar volume is insufficient to accurately describe γ_i^∞ in systems where intermolecular interactions such as hydrogen bonding occur.⁴¹ This accumulates in a significant ARD of $>10^5\%$ in the γ_i^∞ prediction.

Using the HSP to calculate the γ_i^∞ with eq 7 is a significant improvement, compared to the Hildebrand model (eq 5), which is due to taking into account hydrogen bonding and polarity effects.⁸ Still, an ARD of $66.4\% \pm 14.4\%$ is observed for molecular solvents. This may be explained by the inability to differentiate between hydrogen acidity and basicity effects.⁶ Further refining the model using hydrogen acidity and hydrogen basicity, as well as polarizability effects, as taken into account in the MOSCED model, which shows from Figure 2a to be, on average, the most accurate model, with an ARD of $16.2\% \pm 1.35\%$. Unfortunately, no parameters for ILs are available, and therefore this model could not be evaluated for ILs. The three group contribution models that were also evaluated showed comparable ARD values of $32.2\% \pm 1.84\%$, $31.1\% \pm 1.66\%$, and $24.3\% \pm 1.63\%$, respectively, for the modified UNIFAC(Ly), UNIFAC, and modified UNIFAC(Do) for molecular solvents. The ARD of COSMO-RS is within $28.3\% \pm 1.07\%$, which is similar to that observed for the GCM methods. The Abraham model is more accurate than GCM methods and COSMO-RS with an ARD of $21.7\% \pm 1.19\%$. The better accuracy of the Abraham model is due to a

more elaborate description of the various intermolecular interactions via all descriptors.

For ILs, the ARD of all models (see Figure 2b) increase due to the presence of not only dispersion, dipole, and hydrogen bonding interactions, but also ionic interactions between the ionic species and the solute. The ARD of the HSP model is determined to be $168\% \pm 54.5\%$, while the GCM methods perform better with an ARD of $86.2\% \pm 14.6\%$, $86.5\% \pm 15.7\%$, and $122\% \pm 55.9\%$, respectively, for the modified UNIFAC(Ly), UNIFAC, and modified UNIFAC(Do). Although COSMO-RS performed for molecular solvents is comparable to that of the GCMs, for ILs, the ARD is larger ($182\% \pm 16.7\%$). The larger ARD for ILs is known to be (at least partly) caused by neglecting long-range ion–ion interactions and insufficient description of extreme polarization charge densities of ions.¹⁰ The Abraham model performs most accurately for ILs, with an ARD value of $65.1\% \pm 4.50\%$. The accuracy of the Abraham model is interesting for solvent screenings purpose, because of the availability of ion-specific Abraham parameters for 60 cations and 17 anions, allowing rapid assessment of 1020 ILs, since they are binary combinations of these ions. (see the ESI).

The ARDs reported in Figure 2 appear to be larger than errors described in various literature sources.^{6,17–24,42,43,45–48} Similar comparisons have been made by Gmehling et al.,¹⁷ who assessed the accuracy of the UNIFAC models, and Thomas et al.,²⁰ who assessed the accuracy of the MOSCED model. They reported lower ARD values, correspondingly 25.8% (instead of $31.1\% \pm 1.66\%$) and 9.10% (instead of $16.2\% \pm 1.35\%$). Because the error calculation method is the same, and only the used dataset differs, the logical conclusion is that the dataset used in this work includes γ_i^∞ with a higher average error margin. This will be the case for each of the assessed models; hence, these error margins in the dataset will not affect the comparison of the relative accuracies between all models. For comparison with the work of Gmehling et al.,¹⁷ there is a difference in the selected data, because they state that γ_i^∞ values of >100 were excluded, whereas, in this manuscript, these values were included.

Table 1. Accuracy of Eight Predictive Models Differentiated Towards Aliphatic, Aromatic, Halogen, Aprotic Polar, and Protic Polar Compounds in Molecular Solvents^a

Hildebrand Solubility Parameter						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
Solvent	Aliphatic	20,5%	28,3%	>100%	>300%	>10 ⁵ %
	Aromatic	40,3%	17,0%	14,8%	>100%	>500%
	Halogen	56,5%	29,2%	32,5%	73,7%	>100%
	Aprotic polar	>100%	>100%	>100%	>100%	>100%
	Protic polar	>100%	51,6%	45,2%	>100%	67,7%

Hansen Solubility Parameter						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
Solvent	Aliphatic	22,3%	43,3%	44,7%	64,2%	84,5%
	Aromatic	31,1%	18,2%	20,4%	28,7%	75,5%
	Halogen	32,9%	25,1%	30,5%	70,0%	62,4%
	Aprotic polar	47,3%	64,3%	47,1%	39,8%	62,4%
	Protic polar	>100%	57,7%	62,2%	>300%	27,8%

MOSCED Model						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
Solvent	Aliphatic	7,4%	6,7%	7,7%	39,5%	89,1%
	Aromatic	14,3%	19,6%	9,7%	24,2%	>100%
	Halogen	18,1%	9,8%	18,3%	23,2%	37,3%
	Aprotic polar	15,7%	14,8%	15,8%	15,0%	37,3%
	Protic polar	24,6%	16,7%	32,7%	29,8%	19,7%

Abraham Model						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
Solvent	Aliphatic	17,8%	11,8%	6,8%	46,3%	12,5%
	Aromatic	11,8%	20,0%	13,0%	22,9%	27,9%
	Halogen	34,6%	31,8%	16,1%	34,9%	31,7%
	Aprotic polar	21,2%	21,2%	21,7%	22,4%	31,7%
	Protic polar	20,6%	17,5%	22,3%	39,5%	34,1%

COSMO-RS						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
Solvent	Aliphatic	8,6%	15,4%	20,4%	35,6%	45,2%
	Aromatic	10,7%	13,2%	26,7%	24,4%	56,9%
	Halogen	27,6%	36,6%	15,7%	31,4%	33,4%
	Aprotic polar	32,7%	34,4%	32,5%	19,9%	33,4%
	Protic polar	36,3%	32,0%	35,0%	38,1%	23,6%

UNIFAC						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
Solvent	Aliphatic	12,4%	11,0%	26,6%	47,6%	45,6%
	Aromatic	16,6%	18,9%	20,9%	32,4%	24,9%
	Halogen	34,8%	19,4%	18,0%	25,6%	38,4%
	Aprotic polar	40,3%	23,7%	23,0%	27,9%	38,4%
	Protic polar	35,8%	28,3%	32,7%	36,3%	23,4%

mod. UNIFAC (Ly)						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
Solvent	Aliphatic	9,4%	11,9%	21,7%	47,9%	43,6%
	Aromatic	24,9%	19,4%	23,9%	27,3%	43,1%
	Halogen	34,0%	24,7%	19,9%	29,1%	43,0%
	Aprotic polar	41,0%	20,8%	23,7%	23,0%	43,0%
	Protic polar	39,3%	41,3%	34,2%	37,5%	16,8%

mod. UNIFAC (Do)						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
Solvent	Aliphatic	7,1%	5,6%	10,6%	56,8%	18,3%
	Aromatic	17,6%	17,0%	17,1%	26,2%	47,8%
	Halogen	32,0%	13,5%	17,5%	27,9%	35,3%
	Aprotic polar	30,1%	17,4%	25,9%	23,6%	35,3%
	Protic polar	23,2%	18,9%	28,2%	43,4%	28,5%

^aAll 25 binary solute-solvent combinations have been made at 298.15 K. The colors are indicative: white, ARD < 100%; light gray, 100% < ARD < 300%; medium gray, 300% < ARD < 500%; dark gray, 500% < ARD < 10³%; and black, ARD > 10⁵%.

4.2. Molecular Solvents. Although from the averaged model predictions, one general guideline for model selection can be distilled, a more-detailed analysis of subgroups of solutes and solvents allows one to provide a more-sophisticated directive to the use of thermodynamic models for the prediction of γ_i^∞ for various specific chemical families. To this end, all solvents and solutes were classified into five categories, i.e., aliphatic compounds, aromatic compounds, compounds containing a halogen atom, polar aprotic compounds and polar protic compounds, and the model accuracies were evaluated per combination of solvent and solute class.

In Table 1, all model evaluations are shown per combination of solvent and solute categories. Comparison of the Hildebrand, HSP, and MOSCED models clearly shows that models with increasing complexity, i.e., taking hydrogen bonding and polarity into account, and describing the

hydrogen bonding basicity and acidity separately, as well as including polarizability, predict γ_i^∞ with increasing accuracy. All of these models show the largest ARD for protic compounds, including amines, alcohols, and aldehydes, which is a logic result due to the number and type of intermolecular interactions occurring in these systems.

There is no single model that predicts γ_i^∞ most accurately for all solvent and solute category combinations. Each of the models, except for the Hildebrand and HSP models, most accurately predicts a category of solute-solvent pairs. COSMO-RS performs best in systems with only (induced) dipole interactions and in the absence of hydrogen bonding formation. When polarity and hydrogen bonding systems are concerned, COSMO-RS becomes less accurate. The MOSCED and Abraham models perform much better in hydrogen bonding systems, because of the multiple parameters that describe these directional interactions. The various

Table 2. Accuracy of Five Predictive Models Differentiated Towards Aliphatic, Aromatic, Halogen, Aprotic Polar, and Protic Polar Compounds in Ionic Liquids with Various Nonfunctionalized and Functionalized Cations^a

COSMO-RS						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
(-)	Imidazolium	198%	273%	291%	115%	217%
	Pyrrolidinium	60%	182%		37%	100%
	Pyridinium	85%	204%		63%	104%
	Phosphonium	218%	127%		46%	16%
	Sulphonium	69%	223%		154%	139%
FIL	Imidazolium	238%	322%	166%	170%	128%
	Pyrrolidinium	112%			53%	
	Pyridinium	194%				
	Morpholinium	67%				

UNIFAC						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
(-)	Imidazolium	64%	60%	59%	44%	156%
	Pyrrolidinium	27%	37%		26%	94%
	Pyridinium	106%	44%		327%	594%
	Phosphonium					
	Sulphonium	30%	118%		29%	94%
FIL	Imidazolium	48%	81%	102%	208%	183%
	Pyrrolidinium	77%				
	Pyridinium	80%				
	Morpholinium					

Mod. UNIFAC (Ly)						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
(-)	Imidazolium	67%	65%	64%	45%	142%
	Pyrrolidinium	32%	34%		29%	89%
	Pyridinium	109%	41%		309%	508%
	Phosphonium	34%	138%		28%	88%
	Sulphonium					
FIL	Imidazolium	51%	90%	113%	189%	164%
	Pyrrolidinium	79%				
	Pyridinium	80%				
	Morpholinium					

Mod. UNIFAC (Do)						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
(-)	Imidazolium	40%	33%	8%	222%	103%
	Pyrrolidinium	62%	66%		27%	53%
	Pyridinium	59%	228%		47%	51%
	Phosphonium		11%			
	Sulphonium	59%	29%	99%	527%	574%
FIL	Imidazolium	28%				
	Pyrrolidinium					
	Pyridinium					
	Morpholinium					

Abraham model						
		Solute				
		Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
(-)	Imidazolium	56%	60%	60%	36%	86%
	Pyrrolidinium	51%	54%		54%	69%
	Pyridinium	51%	48%		27%	30%
	Phosphonium	54%	70%		7%	30%
	Sulphonium					
FIL	Imidazolium	98%	90%	67%	78%	116%
	Pyrrolidinium	30%				
	Pyridinium					
	Morpholinium	19%				

^aAll 25 binary solute–solvent combinations have been made at 298.15 K. Color legend: white, ARD <100%; light gray, 100% < ARD < 300%; medium gray, 300% < ARD < 500%; dark gray, 500% < ARD < 10³%; and black, ARD >10⁵%.

UNIFAC models appear to be most accurate for a few categories of chemical interaction systems, which may arise from the empirical nature of the UNIFAC models based on fitting the model parameters to experimental data. The variation between the UNIFAC models can also arise from the different fitting procedures for the determination of their empirical constants. Finally, the differences in the formulation of their combinatorial term can induce variation in activity coefficient prediction.

4.3. Ionic Liquids. Overall, the ARD in predicted γ_i^∞ in ILs is larger than that observed in molecular solvents. Not only is the additional electrostatic intermolecular interaction of the charges on the ions with the solutes responsible for this, but also the additional competition between the solute-related intermolecular interactions and the interactions between the ions in the IL plays an important role. Furthermore, ILs with a cation containing, besides the central ionic moiety, also a second moiety (e.g., ether, hydroxyl, or unsaturated bond), are collectively evaluated as functionalized ionic liquids (FILs).

4.3.1. Cations. A systematic analysis was made for various classes of cations, and the cation class-specific model

performances are listed in Table 2. A larger ARD is generally obtained for FILs, because of the fact additional intermolecular and intramolecular interactions occur with the moieties present on the cation tails. Also, COSMO-RS clearly has severe difficulties in predicting accurate γ_i^∞ values. Analogous to the trends observed in molecular solvents, the ARD increase from apolar to polar solutes, because of hydrogen bonding effects. Overall, the performance of the Abraham model is superior to that of the other models, although some systems can be more accurately described using a variant of UNIFAC. For instance, the predictions for aliphatic, aromatic, and halogen solutes in imidazolium cations are more accurately predicted with modified UNIFAC(Do). This is most likely due to the large dataset available for imidazolium cations, hence improving the empirical fit of the modified UNIFAC(Do).

4.3.2. Anions. The nature of the anion in ILs has been identified as a key factor determining the γ_i^∞ parameter for solutes;⁴⁹ therefore, it is also important to list the anion-category-specific ARD in the predictions of γ_i^∞ , which is done in Table 3. By combining the information in Table 2 with the information in Table 3, it becomes clear that the large ARD

COSMO-RS					
	Solute				
	Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
[NTf ₂]	159%	211%	163%	154%	162%
[BF ₄]	504%	468%		206%	216%
[SCN]	123%	274%		48%	104%
[CF ₃ SO ₃]	88%	139%		116%	84%
[MDEGSO ₄]	184%	118%	526%	46%	16%
[OSO ₄]	246%	125%	206%		199%
[PF ₆]	208%	371%			9%
[B(CN) ₄]	241%	330%		137%	337%
[TFA]	287%	263%			
[DMPO ₄]	112%			53%	
[DCA]	194%				

UNIFAC				
Solute				
Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
64%	74%	100%	266%	279%
118%	89%		25%	184%
66%	61%		39%	89%
55%	50%		54%	57%
34%	23%	8%		
19%	17%	86%		65%
76%	73%			94%
16%	39%		15%	53%
25%	24%			
77%				
80%				

Mod. UNIFAC (Ly)					
	Solute				
	Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
[NTf ₂]	66%	85%	109%	246%	253%
[BF ₄]	115%	89%		32%	129%
[SCN]	70%	61%		39%	86%
[CF ₃ SO ₃]	58%	51%		48%	57%
[MDEGSO ₄]	45%	40%	24%		
[OSO ₄]	36%	31%	86%		38%
[PF ₆]	80%	77%			93%
[B(CN) ₄]	22%	38%		30%	61%
[TFA]	28%	34%			
[DMPO ₄]	79%				
[DCA]	80%				

Mod. UNIFAC (Do)				
Solute				
Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
48%	35%	88%	442%	453%
66%	38%		163%	136%
	47%		12%	
65%	50%		90%	69%
61%	61%			
15%	28%	11%		50%
34%	16%			
28%				

Abraham model					
	Solute				
	Aliphatic	Aromatic	Halogen	Aprotic polar	Protic polar
[NTf ₂]	80%	102%	64%	68%	114%
[BF ₄]	69%	21%		33%	114%
[SCN]	61%	48%		42%	64%
[CF ₃ SO ₃]	41%	46%		24%	41%
[MDEGSO ₄]			124%		
[OSO ₄]	28%	26%	19%		30%
[PF ₆]	57%	98%			11%
[B(CN) ₄]	52%	57%		17%	56%
[TFA]	61%	54%			
[DMPO ₄]	30%				

Table 4. Overview of the Most Accurate Predictive Model of Each Specific Category of Binary Molecular Solvent–Solute Mixtures

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observed with various UNIFAC models and the Abraham model originate from very large ARD observed for a few anions. ARD values of >100% are caused by the bis-(trifluoromethane)sulfonimide [NTf₂], tetrafluoroborate [BF₄], and diethylene glycolmonomethyl ether sulfate [MDEGSO₄] anions, which indicates that improving these correlations will greatly improve the overall average accuracy of these models. COSMO-RS appears to have difficulties in each anion, indicating that the problems of COSMO-RS do not arise from a particular intermolecular interaction induced by one or more specific anion(s).

5. MODEL SELECTION

In Table 4, for each combination of chemical systems, the model that most accurately predicts for that combination of chemical classes is listed. However, this is not the only selection criteria of importance. Solvation models, such as MOSCED and the Abraham model, can only be used when all molecular parameters are known. GCMs and COSMO-RS are much more flexible, in the sense that (almost) every molecular structure can be drawn and the γ_i^∞ parameter can be predicted. To improve the applicability range of MOSCED and the Abraham model, recent efforts are made to predict the molecular parameters for MOSCED by using a GCM⁵⁰ or by using quantum mechanical charge density calculations.^{51,52} Moreover, isobaric vapor–liquid predictions using MOSCED have been shown to outperform UNIFAC.³² Regarding the Abraham model, it has been shown that solute parameters can be predicted by multilinear regression analysis and computational neural networks.⁵³ The limited temperature range of, often, only 298.15 K is a drawback for SMs and LSER, and attempts have been made toward temperature-independent parameters;⁵⁴ however, this is not generalized yet.

It should be taken into consideration that the accuracy for predicting γ_i^∞ by GCMs can be improved when only γ_i^∞ data are regressed and not the thermodynamic data of other forms (excess enthalpy for example). However, this will reduce the accuracy for other thermodynamic properties.⁴ Therefore, a suggestion could be made to regress a separate γ_i^∞ -specific GCM that can obtain the most accurate γ_i^∞ predictions.

Overall, the model choice for the most accurate γ_i^∞ must be a stepwise procedure. First, it should be ascertained whether all molecular parameters of the chosen molecules are available in the literature. If this is not the case, they may be predicted using theoretical or regression methods, although the accuracy of these methods should always be taken into consideration. When satisfactory molecular parameters are available, the best practice matrix in Table 4 can be used to select the best model. When the molecular parameters are lacking or are questionable (strongly deviating from the same parameters for comparable molecules), then GCMs or COSMO-RS should be used for the prediction. Regarding ILs, the Abrahams model is the most accurate model when the molecular parameters are known. Otherwise, GCMs or COSMO-RS must be used, although this will significantly increase the probability of deviation from the true value.

Lastly, the application should be taken into consideration. GCMs and COSMO-RS are able to predict a much wider range of molecular properties than the SMs and LSER.

6. CONCLUSIONS

Several models have been assessed for their ability to predict the infinite dilution activity coefficient (γ_i^∞) for solutes of various classes, in molecular solvents categorized in the same classes, and in ionic liquids (ILs). A larger ARD was observed for ILs than for molecular solvents, because of the additional ionic interactions. Averaged overall, the MOSCED model was the most accurate model for the prediction of γ_i^∞ of all solute classes in molecular solvents, with an ARD of $16.2\% \pm 1.35\%$. The UNIFAC group contribution methods (GCMs), COSMO-RS, and the Abraham models perform comparably, with ARD values of $24.3\%–32.2\%$. Models using the Hildebrand parameter and the Hansen solubility parameters (HSP) are significantly less accurate, because of an insufficient description of intermolecular interactions such as hydrogen bonds. To predict the γ_i^∞ in ILs, overall, the Abraham model is the most accurate model, with an ARD value of $65.1\% \pm 4.50\%$. The GCMs are less accurate, with ARD values of $86.2\%–122\%$, while COSMO-RS is far less accurate, with an ARD value of $182\% \pm 16.7\%$, because of a deficient description of long-range interactions.

Upon classification of solutes and molecular solvents and evaluating the model prediction accuracy for each of the solvent and solute classes, it is observed that each of the models, except for the Hildebrand parameter and HSP, is most accurate for specific classes of binary solute–solvent pairs, although the accuracy decreases with the polarity of the solute. For ILs, using the overall averages, the Abraham model is most accurate, although several cations are more accurately described with modified UNIFAC(Ly) or modified UNIFAC-(Do). The large ARD values from the UNIFAC models and the Abraham model are mainly due to large ARD values for the [NTf₂], [BF₄], and [MDEGSO₄] anions. Hence, improving the prediction of these anions will greatly increase their overall prediction accuracy. In addition, the most accurate model for molecular solvents (MOSCED) could not be assessed for ILs. Therefore, an extension of MOSCED toward ILs may become an accurate tool in predicting accurate γ_i^∞ values in ILs.

These evaluation results are applicable when each molecular parameter is either known or accurately predicted. If so, the most accurate model for estimation of γ_i^∞ is dependent on both the solute and solvent categories under evaluation. Still, using a model to predict γ_i^∞ for IL screening should be done with caution, since these, on average, easily exceed deviations of 65%.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.9b00727.

Specific parameters, equations, and predictions for the various models; all experimental γ_i^∞ used in the comparison, along with the corresponding references (PDF)

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Notes

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NOMENCLATURE

Abbreviations

AD = average deviation
 ARD = average relative deviation
 UNIQUAC = Universal Quasichemical
 UNIFAC = UNIQUAC Functional-group Activity Coefficients
 COSMO-RS = Conductor like Screening Model for Real Solvents
 DFT = density functional theory
 ESI = electronic Supporting Information
 MOSCED = Modified Separation of Cohesive Energy Density
 ILs = ionic liquids
 FILs = functionalized ionic liquids
 NRTL = nonrandom two-liquid model
 GCM = group contribution method
 HSP = Hansen solubility parameters
 HF = Hartree–Fock
 SCF = self-consistent field
 SM = solvation models
 TZVPD-Fine = fine grid triple- ζ valence polarized basis set
 $[\text{NTf}_2]$ = bis(trifluoromethylsulfonyl)imide
 $[\text{BF}_4]$ = tetrafluoroborate
 $[\text{SCN}]$ = thiocyanate
 $[\text{CF}_3\text{SO}_3]$ = trifluoromethanesulfone
 $[\text{MDEGSO}_4]$ = 2-(2-methoxyethoxy)ethyl sulfate
 $[\text{OSO}_4]$ = octyl sulfate
 $[\text{PF}_6]$ = hexafluorophosphate
 $[\text{B}(\text{CN})_4]$ = tetracyanoborate
 $[\text{TFA}]$ = trifluoroacetate
 $[\text{DMPO}_4]$ = dimethylphosphate
 $[\text{DCA}]$ = dicyanamide

Symbols

γ_i = activity coefficient of compound i
 γ_i^∞ = infinite dilution activity coefficient of compound i
 γ_i^c = combinatorial term of the activity coefficient of compound i
 γ_i^R = residual term of the activity coefficient of compound i
 x_i = molar fraction of compound i
 Φ_i = volume fraction of compound i
 Φ_i = modified volume fraction of compound i , reported by Weidlich et al.⁴
 Φ_i'' = modified volume fraction of compound i , reported by Larsen et al.⁵
 χ_{12} = Flory–Huggins interaction parameter between compounds 1 and 2
 $^i\delta$ = Hildebrand parameter of compound i
 $^i\delta_D$ = Hansen parameter for dispersion interactions of compound i
 $^i\delta_{HB}$ = Hansen parameter for hydrogen bonding interactions of compound i
 $^i\delta_P$ = Hansen parameter for polar interactions of compound i

α = empirical constant added to Hansen solubility parameters, as reported by Lindvig et al.⁹
 c = cohesive pressure or cohesive energy density
 c = fitting constant of the Abraham model
 N = number of data points
 ΔU_{vap} = molar vaporization energy
 ΔH_{vap} = molar vaporization enthalpy
 iV_m = molar volume of compound i
 T = temperature (K)
 R = universal gas constant; $R = 8.3145 \text{ J K}^{-1} \text{ mol}^{-1}$
 $^i\lambda$ = MOSCED dispersion constant for compound i
 $^i\tau$ = MOSCED polarity constant for compound i
 iq = MOSCED induction constant for compound i
 $^i\alpha$ = MOSCED hydrogen bond acidity constant for compound i
 $^i\beta$ = MOSCED hydrogen bond basicity constant for compound i
 $^i\tau^T$ = temperature-corrected MOSCED polarity constant for compound i
 $^i\alpha^T$ = temperature-corrected MOSCED hydrogen bond acidity constant for compound i
 $^i\beta^T$ = temperature-corrected MOSCED hydrogen bond basicity constant for compound i
 $^i\xi$ = MOSCED empirical asymmetric constant for compound i
 $^i\psi$ = MOSCED empirical asymmetric constant for compound i
 θ_i = surface fraction of compound i
 q_i = coordination number of compound i
 $\nu_k^{(i)}$ = occurrence of each moiety k in surrounded by moiety i
 Γ_k = overall activity of moiety k
 $\Gamma_k^{(i)}$ = activity of moiety k solely surrounded by moiety i
 Q_k = van der Waals surface of moiety k
 R_k = van der Waals volume of moiety k
 Ψ_{mk} = group binary interaction parameter between moieties m and k
 $e_{c/a}$ E = Abraham descriptors for σ - and π -electron interactions (subscripts c and a indicate cationic and anionic species, respectively)
 $s_{c/a}$ S = Abraham descriptors for (induced) multipole moments interactions (subscripts c and a indicate cationic and anionic species, respectively)
 $a_{c/a}$ A = Abraham descriptors for tendency to accept hydrogen or donate electron (subscripts c and a indicate cationic and anionic species, respectively)
 $b_{c/a}$ B = Abraham descriptors for tendency to donate hydrogen or accept electron (subscripts c and a indicate cationic and anionic species, respectively)
 $v_{c/a}$ V = Abraham descriptors for the work/tendency required to form cavities (subscripts c and a indicate cationic and anionic species, respectively)
 $l_{c/a}$ L = Abraham descriptors for the work/tendency required to form cavities (subscripts c and a indicate cationic and anionic species, respectively)
 P_i^0 = vapor pressure of the pure solvent
 K_S = gas–solvent partition coefficient
 P_S = water–solvent partition coefficient
 E_{contact} = local contact energy
 E_{misfit} = misfit energy
 E_{HB} = energy due to hydrogen bonding effects
 σ = polarization charge density
 σ' = polarization charge density on the opposite side

REFERENCES

- (1) Kiss, A. A.; et al. Separation technology—Making a difference in biorefineries. *Biomass Bioenergy* **2016**, *95*, 296–309.
- (2) Sholl, D. S.; Lively, R. P. Seven chemical separations to change the world. *Nature* **2016**, *532* (7600), 435.
- (3) Abraham, M. H. Scales of solute hydrogen-bonding: their construction and application to physicochemical and biochemical processes. *Chem. Soc. Rev.* **1993**, *22* (2), 73–83.
- (4) Weidlich, U.; Gmehling, J. A modified UNIFAC model. 1. Prediction of VLE, hE, and γ_{∞} . *Ind. Eng. Chem. Res.* **1987**, *26* (7), 1372–1381.
- (5) Larsen, B. L.; Rasmussen, P.; Fredenslund, A. A modified UNIFAC group-contribution model for prediction of phase equilibria and heats of mixing. *Ind. Eng. Chem. Res.* **1987**, *26* (11), 2274–2286.
- (6) Lazzaroni, M. J.; et al. Revision of MOSCED parameters and extension to solid solubility calculations. *Ind. Eng. Chem. Res.* **2005**, *44* (11), 4075–4083.
- (7) Hildebrand, J. H. *Solubility of Non-electrolytes*; Reinhold Publishing Co.: New York, 1936.
- (8) Hansen, C. M. The three dimensional solubility parameter. *J. Paint Technol.* **1967**, *39*, 105.
- (9) Lindvig, T.; Michelsen, M. L.; Kontogeorgis, G. M. A Flory–Huggins model based on the Hansen solubility parameters. *Fluid Phase Equilib.* **2002**, *203* (1), 247–260.
- (10) Klamt, A.; Eckert, F.; Arlt, W. COSMO-RS: an alternative to simulation for calculating thermodynamic properties of liquid mixtures. *Annu. Rev. Chem. Biomol. Eng.* **2010**, *1*, 101–122.
- (11) Fredenslund, A.; Jones, R. L.; Prausnitz, J. M. Group-contribution estimation of activity coefficients in nonideal liquid mixtures. *AIChE J.* **1975**, *21* (6), 1086–1099.
- (12) Alessi, P.; Fermeglia, M.; Kikic, I. Significance of dilute regions. *Fluid Phase Equilib.* **1991**, *70* (2–3), 239–250.
- (13) London, F. The general theory of molecular forces. *Trans. Faraday Soc.* **1937**, *33*, 8b–26.
- (14) Debye, P. van der Waals' cohesion forces. *Phys. Z.* **1920**, *21*, 178–187.
- (15) Keesom, W. H. Van der Waals attractive force. *Phys. Z.* **1921**, *22*, 129–141.
- (16) Israelachvili, J. N. *Intermolecular and Surface Forces*; Academic Press: Burlington, MA, 2011.
- (17) Gmehling, J.; Li, J.; Schiller, M. A modified UNIFAC model. 2. Present parameter matrix and results for different thermodynamic properties. *Ind. Eng. Chem. Res.* **1993**, *32* (1), 178–193.
- (18) Putnam, R.; et al. Prediction of infinite dilution activity coefficients using COSMO-RS. *Ind. Eng. Chem. Res.* **2003**, *42* (15), 3635–3641.
- (19) Voutsas, E. C.; Tassios, D. P. Prediction of infinite-dilution activity coefficients in binary mixtures with UNIFAC. A critical evaluation. *Ind. Eng. Chem. Res.* **1996**, *35* (4), 1438–1445.
- (20) Thomas, E. R.; Eckert, C. A. Prediction of limiting activity coefficients by a modified separation of cohesive energy density model and UNIFAC. *Ind. Eng. Chem. Process Des. Dev.* **1984**, *23* (2), 194–209.
- (21) Park, J. H.; Carr, P. W. Predictive ability of the MOSCED and UNIFAC activity coefficient estimation methods. *Anal. Chem.* **1987**, *59* (21), 2596–2602.
- (22) Castells, C. B.; et al. Comparative study of semitheoretical models for predicting infinite dilution activity coefficients of alkanes in organic solvents. *Ind. Eng. Chem. Res.* **1999**, *38* (10), 4104–4109.
- (23) Xue, Z.; Mu, T.; Gmehling, J. Comparison of the a priori COSMO-RS models and group contribution methods: original UNIFAC, modified UNIFAC (Do), and modified UNIFAC (Do) consortium. *Ind. Eng. Chem. Res.* **2012**, *51* (36), 11809–11817.
- (24) Hait, M. J.; et al. Space predictor for infinite dilution activity coefficients. *Ind. Eng. Chem. Res.* **1993**, *32* (11), 2905–2914.
- (25) Flory, P. J. Thermodynamics of high polymer solutions. *J. Chem. Phys.* **1942**, *10* (1), 51–61.
- (26) Huggins, M. L. Some properties of solutions of long-chain compounds. *J. Phys. Chem.* **1942**, *46* (1), 151–158.
- (27) Sayegh, S.; Vera, J. Lattice-model expressions for the combinatorial entropy of liquid mixtures: a critical discussion. *Chemical Engineering Journal* **1980**, *19* (1), 1–10.
- (28) Krooshof, G. J.; Tuinier, R.; de With, G. Generalization of Guggenheim's combinatorial activity coefficient equation. *J. Mol. Liq.* **2018**, *266*, 467–471.
- (29) Kikic, I.; et al. On the combinatorial part of the UNIFAC and UNIQUAC models. *Can. J. Chem. Eng.* **1980**, *58* (2), 253–258.
- (30) Barton, A. F. *CRC Handbook of Solubility Parameters and Other Cohesion Parameters*; CRC Press: Boca Raton, FL, 1991.
- (31) Hansen, C. *Hansen Solubility Parameters: A User's Handbook*; CRC Press: Boca Raton, FL, 2000; p 168.
- (32) Dhakal, P.; et al. Application of MOSCED To Predict Limiting Activity Coefficients, Hydration Free Energies, Henry's Constants, Octanol/Water Partition Coefficients, and Isobaric Azeotropic Vapor–Liquid Equilibrium. *J. Chem. Eng. Data* **2018**, *63* (2), 352–364.
- (33) Staverman, A. The entropy of high polymer solutions. Generalization of formulae. *Recueil des Travaux Chimiques des Pays-Bas* **1950**, *69* (2), 163–174.
- (34) Abboud, J. L.; Kamlet, M. J.; Taft, R. Regarding a generalized scale of solvent polarities. *J. Am. Chem. Soc.* **1977**, *99* (25), 8325–8327.
- (35) Kamlet, M. J.; Abboud, J. L.; Taft, R. The solvatochromic comparison method. 6. The π^* scale of solvent polarities. *J. Am. Chem. Soc.* **1977**, *99* (18), 6027–6038.
- (36) Kamlet, M. J.; et al. Linear solvation energy relationships. 13. Relationship between the Hildebrand solubility parameter, δ , and the solvatochromic parameter, π . *J. Am. Chem. Soc.* **1981**, *103* (20), 6062–6066.
- (37) Kamlet, M. J.; Taft, R. The solvatochromic comparison method. I. The β -scale of solvent hydrogen-bond acceptor (HBA) basicities. *J. Am. Chem. Soc.* **1976**, *98* (2), 377–383.
- (38) Taft, R.; Abboud, J.-L. M.; Kamlet, M. J. Solvatochromic comparison method. 20. Linear solvation energy relationships. 12. The δ term in the solvatochromic equations. *J. Am. Chem. Soc.* **1981**, *103* (5), 1080–1086.
- (39) Sprunger, L.; et al. Characterization of room-temperature ionic liquids by the Abraham model with cation-specific and anion-specific equation coefficients. *J. Chem. Inf. Model.* **2007**, *47* (3), 1123–1129.
- (40) Eckert, F.; Klamt, A. Fast solvent screening via quantum chemistry: COSMO-RS approach. *AIChE J.* **2002**, *48* (2), 369–385.
- (41) Hildebrand, J.; Scott, R. *The Solubility of Nonelectrolytes*, 3rd Edition; Reinhold Publishing Co.: New York, 1950.
- (42) Wang, J.; et al. Correlation of infinite dilution activity coefficient of solute in ionic liquid using UNIFAC model. *Fluid Phase Equilib.* **2008**, *264* (1–2), 235–241.
- (43) Lei, Z.; et al. Extension of the UNIFAC model for ionic liquids. *Ind. Eng. Chem. Res.* **2012**, *51* (37), 12135–12144.
- (44) Kato, R.; Gmehling, J. Systems with ionic liquids: Measurement of VLE and γ_{∞} data and prediction of their thermodynamic behavior using original UNIFAC, mod. UNIFAC (Do) and COSMO-RS (Ol). *J. Chem. Thermodyn.* **2005**, *37* (6), 603–619.
- (45) Nebig, S.; Liebert, V.; Gmehling, J. Measurement and prediction of activity coefficients at infinite dilution (γ_{∞}), vapor–liquid equilibria (VLE) and excess enthalpies (H E) of binary systems with 1, 1-dialkyl-pyrrolidinium bis (trifluoromethylsulfonyl) imide using mod. UNIFAC (Dortmund). *Fluid Phase Equilib.* **2009**, *277* (1), 61–67.
- (46) Nebig, S.; Böls, R.; Gmehling, J. Measurement of vapor–liquid equilibria (VLE) and excess enthalpies (H E) of binary systems with 1-alkyl-3-methylimidazolium bis (trifluoromethylsulfonyl) imide and prediction of these properties and γ_{∞} using modified UNIFAC (Dortmund). *Fluid Phase Equilib.* **2007**, *258* (2), 168–178.
- (47) Paduszyński, K.; Domańska, U. Extension of modified UNIFAC (Dortmund) matrix to piperidinium ionic liquids. *Fluid Phase Equilib.* **2013**, *353*, 115–120.
- (48) Nebig, S.; Gmehling, J. Prediction of phase equilibria and excess properties for systems with ionic liquids using modified

UNIFAC: Typical results and present status of the modified UNIFAC matrix for ionic liquids. *Fluid Phase Equilib.* **2011**, 302 (1–2), 220–225.

(49) Wytze Meindersma, G.; Podt, A. J.; de Haan, A. B. Selection of ionic liquids for the extraction of aromatic hydrocarbons from aromatic/aliphatic mixtures. *Fuel Process. Technol.* **2005**, 87 (1), 59–70.

(50) Dhakal, P.; et al. GC-MOSCED: A group contribution method for predicting MOSCED parameters with application to limiting activity coefficients in water and octanol/water partition coefficients. *Fluid Phase Equilib.* **2018**, 470, 232–240.

(51) Diaz-Rodriguez, S.; et al. Predicting cyclohexane/water distribution coefficients for the SAMPL5 challenge using MOSCED and the SMD solvation model. *J. Comput.-Aided Mol. Des.* **2016**, 30 (11), 1007–1017.

(52) Phifer, J. R.; et al. Computing MOSCED parameters of nonelectrolyte solids with electronic structure methods in SMD and SM8 continuum solvents. *AIChE J.* **2017**, 63 (2), 781–791.

(53) Jover, J.; Bosque, R.; Sales, J. Determination of Abraham Solute Parameters from Molecular Structure. *J. Chem. Inf. Comput. Sci.* **2004**, 44 (3), 1098–1106.

(54) Mintz, C.; et al. Characterization of the Partitioning of Gaseous Solutes Into Humic Acid with the Abraham Model and Temperature-Independent Equation Coefficients. *QSAR Comb. Sci.* **2008**, 27 (4), 483–491.